New Attempt

- Due Nov 20, 2024 by 2:59am
- Points 35
- Submitting a text entry box, a website url, or a file upload
- Robot Project

Robot Project Challenge 3

- **Overview:**

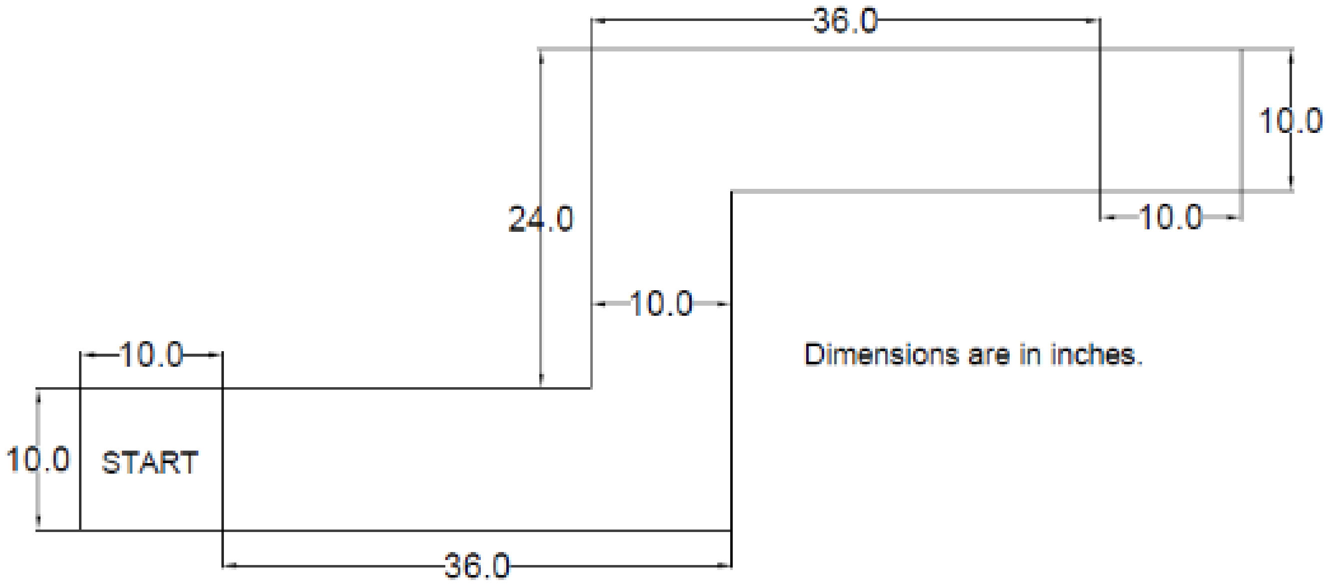
This is the third robot challenge! Begin by meeting with your team to discuss the successes and challenges faced in Challenge 2. Together, program your robot to navigate a specified maze using the distance sensor to avoid obstacles. Be sure to record each step of the process and document all relevant details in your digital design notebook.

Reminder: It's crucial to dedicate time to your **Individual laser cutting project (https://northeastern.instructure.com/courses/189179/assignments/2352195)** and incorporate the best design of your team into your robot, ideally by **Challenge 3.**

- **Tasks:**
  - **Task 1: Reflection: As a team, reflect on the followings and record your insights in your design notebook:**
    - Review the successes and failures encountered during Challenge 2, and reflect on what you learned from each aspect.
    - Reflect on your teamwork experience during Challenge 2. Are there any concerns about the level of effort or time each member contributed? Are you satisfied with the quality of work produced by both your teammates and yourself? Based on these reflections, set specific goals to improve teamwork and team dynamics for the next challenge.
  - **Task 2**: **Program the robot to drive autonomously in an known maze made of carboard that looks exactly like Challenge 2 path, avoiding walls using sensors. You should NOT pre-program your robot.**
    - Use button/switch/some other trigger to begin the program.
    - Operated **autonomously** using sensors included in the Sparkfun kit.
    - If you choose to use the distance sensor, note that you have a 3D printed mount to secure the sensor on your robot. You'd need a F to M jumper wire to connect the sensor to your boards.
    - Able to drive
    - Able to turn

- Able to back up if needed.
- Able to travel within the 10" boundary without hitting any walls
- Stops at the end of the course (there will be a wall at the end of the maze)
- Look at the code in Projects 5B and 5C and think about how you will program your robot to do this.
- I suggest recording all of your attempts on video and submitting the best one.

Robot Course:



- **Task 3: Design your Modular Maze in AutoCAD.**
  - The modular maze designed as part of challenge 3 and built later by each team will be used in the Challenge 4 demo to test how effectively the robot can autonomously navigate an unknown maze.
  - Your completed cardboard maze, ready for another team to use with their robot, due by 12.3.24 in class.
  - **Requirements for the Modular Maze**
    - At least three turns (and no more than 5)
    - Walls at least 6 inches tall
    - Able to be sensed by the sensors included in the Sparkfun kit
    - Can be reconfigured quickly (less than 2 minutes)
    - Constraint: Narrowest point at least 10 inches wide (*match the width of the maze from Challenge 2)*
  - As a team, choose one of the idea generation techniques from the chapter on Tophat and apply it to design the Modular Maze part of the Robot Project.
  - Then, use KTDA technique to quantitatively compare the solutions with respect to the goals.

- Then for your top choice create a dimensioned top view drawing in AutoCAD. Fit your drawing on a title block that includes names of all team members. You can use the title block template you made in class as part of AC-3.
- Add a note to the notes section of your drawing about the height of the walls and the drawing units.
- **Include in your design notebook:** Evidence of ideation (sketches, etc), Evidence of decision-making, a screenshot of dimensioned drawing made in AutoCAD

- **Task 4: Record everything in your digital design notebook and share the link with Professor. Make sure the link is accessible and does not require permission. You can test this by copy and pasting the link to an incognito window.**
  - Your reflections from task 1
  - Flowchart your algorithm for programming the robot using draw.io and submit
  - Record a video of your best attempt (can be from during class or out of class)
    - You can include a link to the video on YouTube, Google drive, or other cloud service if you can't/don't want to upload the actual video
  - Plenty of pictures
  - Meeting minutes
  - Any success or failure
  - Be sure to update your digital design notebook every time you work on your robot project individually or as a team.
  - Modular maze evidence: A short description of your design considerations and choices (1-2 paragraphs)
    - Evidence of ideation (sketches, etc)
    - Evidence of decision-making
    - A dimensioned drawing made in AutoCAD of your maze
    - Fit your drawing on a title block that includes names of all team members and notes requested

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File Submission

As a group, submit the followings:

- Provide an accessible link to your digital design notebook.
- Arduino code

Robot Challenge 3

Criteria	Ratings		Pts
Robot Success (in-person or video) 3 pts each (6 requirements: uses a trigger to start the program, autonomous, makes turns, drives straight, stops at the end, does not hit the walls)	**20 pts** **Full Marks**	**0 pts** **No Marks**	20 pts
Code uploaded and compiles	**5 pts** **Full Marks**	**0 pts** **Does not compile or has errors**	5 pts
Digital Design Notebook Details Has all of the details requested	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 pts
Modular Maze Design Has all the details requested	**5 pts** **Full Marks**	**0 pts** **No Marks**	5 pts
			Total Points: 35